

3-Day Task Plan: OCP MXFP8 (E4M3) Dot Product in Python

Goal: build, test and explain a Python model of the OCP MXFP8 E4M3 block and general dot product.

FORMAT	E4M3 (1S/4E/3M)	SCALE	E8M0 shared scale	BLOCK	32 elements	CORE EQUATION	$\text{Dot} = X_A \times X_B \times \sum P_A[i]P_B[i]$
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Daily task layout

Day / Focus	Objective	Tasks	Deliverable	Checkpoint
DAY 1 E4M3 foundation	Understand the E4M3 encoding and build a working element-level model.	1. Review E4M3: 1 sign, 4 exponent, 3 fraction bits; bias 7. 2. Implement decode_e4m3(code). 3. Enumerate all 256 encodings; identify finite values and NaN. 4. Implement quantize_e4m3(x) with round-to-nearest, ties-to-even. 5. Test 0, ±1, 0.5, 3.14, max finite and subnormal values.	e4m3.py containing decode, quantize and basic tests.	Can explain why an FP32 value changes when quantized to E4M3.
DAY 2 One MX block	Add OCP microscaling and implement one 32-element MXFP8 block dot product.	1. Understand E8M0 as a power-of-two shared scale. 2. Implement choose_scale(values) for a 32-value block. 3. Implement quantize_mx_block(): scale → E4M3 quantization. 4. Implement reconstruct_mx_block(). 5. Implement mx_dot_block(A,B) = $X_A \times X_B \times \sum P_A[i]P_B[i]$.	mx.py accepts two 32-element vectors and returns their MXFP8 block dot product.	Can draw the data flow: FP values → scale → E4M3 → dot product.
DAY 3 General dot + verification	Extend to arbitrary vector lengths and quantify the error against FP32.	1. Pad inputs to a multiple of 32 and split into blocks. 2. Implement mx_dot(A,B) as the sum of block dot products. 3. Compare against NumPy FP32 dot product. 4. Test: all ones, zeros, mixed signs, mixed magnitudes and random vectors. 5. Report absolute/relative error; optional scatter plot of FP32 vs MXFP8 results.	Final demo plus a short result table showing MXFP8 vs FP32.	Can explain that the error comes mainly from input quantization and block scaling.

Completion criteria

1	E4M3 encode/decode + quantization works	2	32-element MX block scaling works	3	General MX dot product works	4	MX result is compared and explained vs FP32
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Scope rule: no FP4, neural network or RTL. Focus only on E4M3 quantization, E8M0 block scaling, the MX dot product and numerical verification.